

Annual Report to the Executive Directors on the Special Programme for
Research and Training in Tropical Diseases (TDR)

1. This is the second annual report to the Executive Directors. It will concentrate on current developments in scientific output, shortage of funds, and the evolving role of the Bank.

History of TDR

2. The history of TDR was traced in the first annual report to the Executive Directors in July 1982. In brief, the Programme was founded in 1977 by WHO to attract funds and scientific effort to the solution of outstanding problems in the control of the six major tropical diseases: malaria, schistosomiasis, leishmaniasis, filariasis, trypanosomiasis and leprosy. There is a particular focus on the problems of sub-Saharan Africa. The ultimate objective is to produce effective and acceptable tools to control these diseases and to do so by two lines of action: by direct support of fundamental and applied research and by strengthening research capability in endemic countries through institution building and training. Each disease constitutes a public health problem for which there is no control technology currently available to the tropical developing countries which is simple and cheap enough in application. Research on these diseases had been largely neglected by the scientific community and the pharmaceutical industry, but the scientific techniques in immunology and molecular biology developed to combat the diseases of the industrialized countries are potentially powerful new tools in the battle against these developing country scourges.

3. The Bank became a non-contributing sponsor (with WHO and UNDP) of TDR in 1977. Our support was considered to be important in gaining the commitment and participation of multilateral and bilateral donors and the involvement of developing countries. Substantial, assured, long-term support is needed to stimulate promising lines of research by the world's health science research community, to coordinate succeeding experimental and trial stages, and to prepare developing countries to adapt technology and implement programs. The Bank's involvement in TDR complements its health lending which is directed to expanding and strengthening basic health services to bring available technology to people who presently have no access to such services.

4. Apart from establishing the confidence of the donor community through its support of TDR, the Bank's role has been as a sponsor (and member of the Joint Coordinating Board (JCB) and its Standing Committee) to influence the management of the Programme using its experience in the operation and management of development activities; WHO as Executing Agency has been very responsive to such guidance, which in the formative days of TDR led to substantive improvements in its organization. Lastly, the Bank has provided financial management for a fund through which many of the donors prefer to channel their TDR contributions. Although initially the Bank did not itself contribute funds to TDR, in 1981 it recognized that its role would be enhanced if it did so and it decided in that year to contribute US\$2.48 million, which was 10% of the total contributions expected in that calendar year. The figure of 10% was based on the level of Bank participation in CGIAR and the Onchocerciasis Control Programme.

5. In 1980 a major (quinquennial) review of the Programme was made by an independent team of experts. The report of that body concluded that the Programme was effective, well targetted, and well managed both scientifically and administratively and that it was making major contributions towards the solution of outstanding problems in tropical disease control. At its July 1982 annual meeting, the JCB of TDR accepted the report and adopted its recommendations to streamline research management and to consolidate administration.

Recent scientific progress in the Programme of Research and Development

TDR has invested heavily in applying new techniques in cellular biology and social science to fundamental problems of parasite, vector and host. This investment is now paying off, and the past year has seen a sharp increase in the publication of research findings, each making some advance towards disease control. The most striking advances are in immunology, leading to the possibility of vaccines, parasite metabolism leading to new drugs, and vector biology leading to possibilities of new control mechanisms. The number of leads now available are more than the Programme can follow, and therein lies its main immediate problem of obtaining enough funds to sustain its momentum.

Although research progress has been made in all the six diseases, that in malaria and leprosy is perhaps the most striking:

Malaria has increased since 1979, and it is clear that control programs will not eradicate it. Moreover, both parasite and mosquito are becoming resistant to standard drugs and insecticide. TDR has led much of the research that now promises new solutions. A few years ago, a vaccine for malaria seemed barely feasible, but work supported by TDR has now brought the possibility of a vaccine very close to reality, to the stage of producing pure antigen by modern techniques of cell biology. A number of major research institutions with strong financial support have now joined TDR to take this research through the next, expensive development stage, and so to completion and general application.

To combat growing drug resistance TDR has taken a lead in stimulating research institutes and industry to develop a major new drug, mefloquine, and to work on derivatives of an old Chinese drug, Qinghaosu. Resistance to mefloquine could also arise, and Hoffman-Laroche, who will start distributing the drug in 1983, have offered, in the public interest, to keep its release within guidelines agreed with WHO which are designed to delay as long as practical the build-up of resistance; this agreement represents the first such global action with respect to a new drug.

To deal with resistance of mosquitoes to DDT, TDR has advanced the development of alternative methods of vector control, especially the use of bacteria toxic to larvae *B. Thuringiensis* (now being applied operationally in vector control in the Onchocerciasis Control Programme in West Africa) and *B. Sphaericus*, which is now under early field trial. To pursue vector control more firmly, TDR will set up a program of research into basic vector biology.

Leprosy research has made good progress in developing new drugs, understanding the mechanisms of drug resistance, establishing methods of epidemiological study and field trials. But the most encouraging progress is in the preparation of the first batch of killed bacteria suitable for trial as a vaccine. Trials will start next year and, because of the slow rate of development of leprosy, will need at least ten years to complete, one fact that highlights the need for consistent and predictable funding support.

Recent Progress in the Programme to Strengthen Research Capability

A major objective of TDR is to strengthen research and training capability within endemic countries, and one-third of its funds have been committed in this direction. As a measure of its success, 21 of 32 institutions that have received long-term support now receive competitive grants from the Scientific Research Programme; one-third of all papers in the Sixth Report came from scientists in the developing countries; and 35 of the 113 scientists who were awarded training support received all or part of that training in another developing country. TDR now supports eight M.Sc. courses in developing countries.

Financial Support for TDR

Now that the scientific program is manifestly productive and on a firm footing, the main determinant of progress will be the adequacy of funding. Donor support has not increased as projected, contributions having levelled off since 1979, thus curtailing the expansion of the Programme that had been planned. There is now the prospect of important scientific results having to sit on the shelf because funding is not available for their development. The table shows this levelling off of contributions in nominal terms and a fall in terms of real value.

	<u>Contributions US\$ Million equivalent</u>	
	<u>Nominal (US\$)</u>	<u>Real terms (1979 US\$) (WHO estimate)</u>
1979	21.7	21.7
1980	22.7	20.0
1981	24.4	19.0
1982	23.6	17.0

In 1981, the JCB projected a budget for the 1984-85 biennium of US\$70.6 million but, taking into account the probable shortfall of funds, the biennial budget actually recommended by the Scientific and Technical Advisory Committee (STAC) for JCB approval in July 1983 was \$66.7 million. Actual pledges so far for 1984-85 amount only to approximately \$58 million despite the enthusiastic endorsement of the Programme by nearly all members and observers at the recent JCB. This shortfall of funds comes at a particularly unfortunate time, for the programme is now facing major challenges which compete for TDR support. These challenges are consequences of the progress made by TDR so far, and include field trials of vaccines and drugs and the follow-up of new discoveries. Field trials are a particularly costly phase; as an example, trials for leprosy vaccine

have been conservatively costed at \$10 million for the first five years, with expenditure to continue for five or more years thereafter. While TDR cannot provide for all follow-up research and must make judgments on which to choose as the most promising, the risk that potentially important leads will be dropped is obviously increased if program funding is curtailed.

Future Challenges for TDR

Whilst the broad objective remains the identification of solutions to the six chosen diseases, the immediate challenges are:

1. To focus research support on the most promising leads. Here there is an embarrassment of promising leads and increased support for one activity can only curtail another. STAC has proposed increased support for malaria, leprosy, and social and economic research, a main object of this last being to attract economists to the study of tropical disease. TDR management has also to ensure that the balance of the program is maintained when funding problems force cuts;
2. To set up and fund expensive large scale field trials for vaccines and drugs;
3. To consolidate research capability in endemic countries and to transfer responsibility to national governments;
4. To transfer new technologies to national health services;
5. To strengthen co-operation with other institutions and funding agencies and with industry (already there are co-operative programs with 41 industries in 10 states); and
6. To attract increased financial contributions.

Fund raising and the role of the Bank

The Bank's initial decision to help fund TDR was taken both in recognition of the intrinsic worth of the programme and as an encouragement to other donors. The Bank also recognizes that TDR's own funding has a multiplier effect by directing the resources of research institutions and other funding agencies towards tropical diseases.

The Bank's contribution to TDR since FY82 has been at the rate of about 10% of total contributions; in FY84 it has agreed to contribute 12% up to a maximum of \$3.2 million. It has been agreed among the Co-sponsoring Agencies that they should take joint action in attracting contributions and detailed targets and strategies are now to be worked out.

In addition to its direct financial contribution, the Bank hopes to support to a greater extent than hitherto TDR's research activities involving economic and behavioral issues where the Bank has a comparative advantage. Priority areas appear to include further analysis into the economic and human costs of tropical diseases, in order to improve judgements about the importance of tropical disease research generally and

the priorities within it. (This would inter alia point up the case for increased support of TDR.) A related area in which closer collaboration is planned is the analysis of health ministry budgets in countries with major tropical disease problems in order to improve the allocation of resources, in light of the severity of the diseases and the costs of interventions.

NOTE: A technical summary of the Sixth Annual Report of TDR is available on request from the Director, PHN.

TDR: DONOR CONTRIBUTIONS AND ESTIMATES (US\$)

Government/ Organization	Actual Contributions 1982	Cumulative Total to 12/31/1982	Estimate 1983 1/	Estimate 1984 1/
African Dev' Bank		750,000		250,000
Australia	389,970	1,457,217	690,780	800,000
Austria	28,249	431,434	16,575	30,000
Bahamas		500		
Bayer AG	10,000	10,000		
Belgium	321,839	3,107,188	304,859	350,000
Brazil	20,000	40,000	20,000	20,000
Canada	816,727	3,579,186	1,058,708	900,000
China		50,000	50,000	50,000
Cuba	2,500	8,589	2,000	2,500
Cyprus		239		
Denmark	3,140,545	22,628,995	3,000,000	3,000,000
Finland	204,545	808,213	220,183	250,000
France	332,226	975,899	300,000	300,000
Germany, Fed.Rep. of	1,165,938	4,775,439	912,374	500,000
India		127,469	50,000	25,000
IDRC	177,971	1,171,837	100,000	100,000
ILEP	105,229	568,515	99,415	100,000
Iraq		5,000		
Japan	100,000	100,000	100,000	200,000
JSIF	259,851	3,390,767	250,000	250,000
LEP Trust Board, New Zealand		44,576		
Mexico	9,953	9,953	10,000	10,000
Netherlands	717,682	5,959,141	731,025	750,000
Niger		2,252		
Nigeria		421,389	70,000	70,000
Norway	1,108,742	6,047,797	1,790,318	1,800,000
PAHEF			5,000	
Romania		1,995		
SANOFI			37,500	
Sweden	2,500,000	14,752,162	1,699,673	1,700,000
Switzerland	786,164	4,248,188	835,921	800,000
Thrasher Research Fund	10,000	10,000	10,000	10,000
United Kingdom		4,849,938	160,000	200,000
United Rep. of Cameroon		6,613	4,000	4,000
U.S.A.	5,030,000	15,402,912	2,000,000	3,000,000
Wellcome Trust		25,000		
UNDP	1,840,600	9,260,408	2,317,610	2,200,000
World Bank	2,400,000	4,880,000	2,500,000	3,200,000
WHO	1,050,000	7,651,500	1,050,000	1,282,000
Miscellaneous	3,500	14,011	502,250	1,282,000
Total:	<u>22,532,231</u>	<u>117,574,316</u>	<u>22,149,691</u>	<u>23,153,500</u>

Note: FY corresponds with CY

1/ Estimates by TDR staff as of August 31, 1983.